

Questions by topic: Forest resources & deforestation - Set 1 (55 marks)

Answer all questions in the spaces provided.

01 . 1 Describe the resources obtained from forests. Give **one** example for each category. [5 marks]

Timber

Fuel

Food

Fibres

Medicines

01 . 2 Explain why forests are an important carbon reservoir and how they can be used to counteract climate change. [3 marks]

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01 . 3 Explain how forests regulate the hydrological cycle and local climate. [2 marks]

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0 2

Table 1 compares an indigenous mixed broadleaf woodland with a Sitka spruce plantation of the same area in Scotland.

Table 1

Feature	Mixed indigenous woodland	Sitka spruce plantation
Tree species	12	1
Age structure	All ages, dead wood present	All planted in the same year
Planting density	Natural, variable	Close, 2 m spacing
Timber yield / $\text{m}^3 \text{ ha}^{-1} \text{ yr}^{-1}$	4	16
Bird species	38	9
Ground flora species	60	4

0 2

. 1

Use Table 1 to explain why the plantation has a much higher timber yield but far lower biodiversity.

[6 marks]

0 2 . 2 Explain why commercial plantations are felled before the trees reach full size. **[2 marks]**

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0 2 . 3 Explain why there is little demand today for coppiced or pollarded wood. **[2 marks]**

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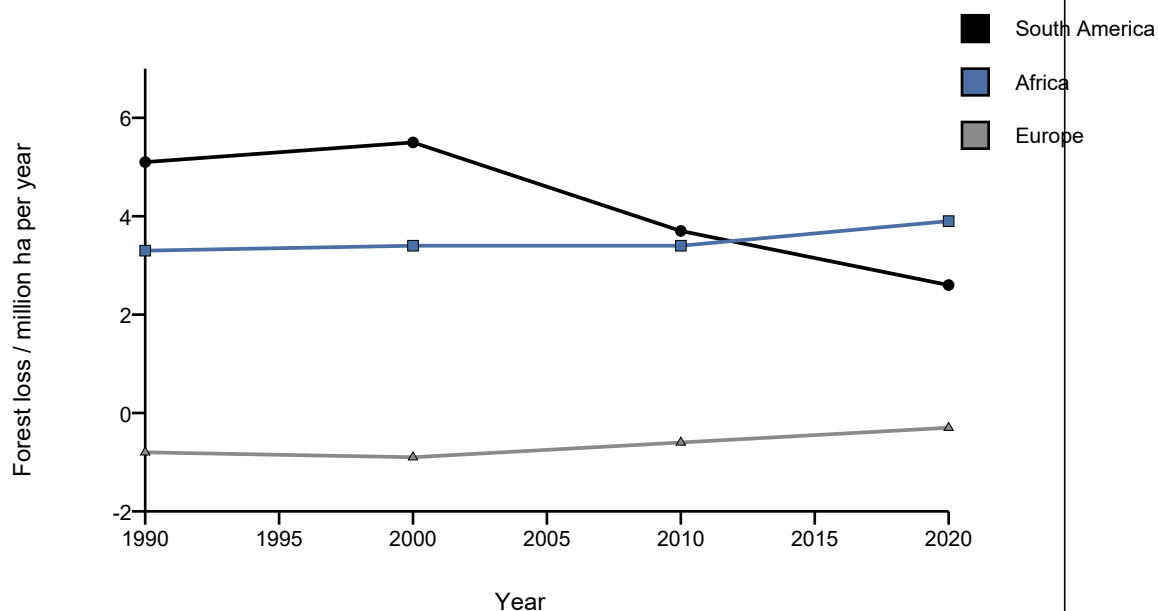
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0 3 Figure 1 shows the area of forest lost each year in three regions between 1990 and 2020.

Figure 1



0 3 . 1 Describe the trends shown in Figure 1. **[3 marks]**

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03 . 2

Explain the main causes of forest loss in South America and Africa.

[4 marks]

03 . 3

Explain why roads and railways cause more forest loss than the area they occupy.

[2 marks]

03 . 4

Explain why softwood from Scandinavia and Canada is generally produced more sustainably than tropical hardwood.

[1 mark]

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0 4 . 1

Explain the effects of deforestation on soil.

[4 marks]

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0 4 . 2

Explain how deforestation affects hydrology and climate.

[4 marks]

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Give **two** ecological impacts of deforestation other than the direct loss of species.

[2 marks]

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The Rio Bravo Conservation and Management Area in Belize exploits mahogany under a sustainable management plan while conserving a rainforest of over 200 tree, 400 bird and 70 mammal species.

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Describe **five** features of the timber harvesting at Rio Bravo that make it sustainable.

[5 marks]

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Explain why involving local communities is essential to the success of the reserve.

[3 marks]

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Explain why selective logging is better for wildlife than clear-felling, and give **one** disadvantage.

[2 marks]

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06 . 1

Which of the following features of a plantation would most increase its wildlife value? Tick one box.

[1 mark]

A Close planting of a single non-indigenous species

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B Mixed indigenous species with a mixed age structure

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C Clear-felling on a 40-year rotation

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D Removal of dead wood

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06 . 2

Describe **four** features of sustainably managed forests and explain how each benefits biodiversity or productivity.

[4 marks]

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END OF QUESTIONS